Faculty of Mathematics and Computer Science
Artificial Intelligence Group

Basic Possibilistic Logic: Examples and Visualization

Seminar Paper
submitted by
Alexander Köhn

Abstract

Possibilistic logic is a non-classical logic framework designed to handle uncertainty by incorporating degrees of possibility. This paper provides a general introduction to possibilistic logic by using strong examples of the apllication.

1 Introduction

Reasoning under uncertainty is a fundamental problem in artificial intelligence. Classical logic, while expressive and well-structured, lacks the ability to handle incomplete or uncertain information effectively. Several approaches have been proposed to address this issue, including possibility theory [4], fuzzy logic [5], and possibilistic logic [1]. The foundation of possibility theory was laid by Zadeh in 1978 [6] as an extension of fuzzy set theory [5], aiming to provide a qualitative framework for handling uncertain information. Building upon Zadeh's work, Dubois and Prade formulated possibilistic logic [1] as a means to integrate possibility theory into formal logic systems.
Possibilistic logic has been further developed for diverse areas such as knowledge representation, database systems, and reasoning under inconsistency. The goal of this paper is to introduce basic possibilistic logic with examples and visualization. The focus of this paper is on introducing common and new examples with more explanation and visualization. In this paper we will only focus on basic possibilistic logic.
This work is structured as follows. In Section 2 basic possibilistic logic is explained with its preliminaries for understanding the syntax and semantics of the logic. Before the core part of this paper the Section 3 compares possibilistic logic with classical logic and the Section 4 relates possibilistic logic with artificial intelligence. The Section 5 shows the use of basic possibilistic logic with multiple examples and graphical explanations. The last Section 6 discusses the results of the previous sections.

2 Fundamentals of Basic Possibilistic Logic

The following introduction on basic possibilistic logic is inspired by the presentation of Dubois and Prade [2]. This section gives a short overview of basic possibilistic logic. For more information, the work [2] is referred.
In this section we will first introduce main concepts of possibility theory. These concepts are necessary to understand possibilistic logic. Going further basic possibilistic logic is introduced with the syntax, axioms, inference rules and semantic aspects.

2.1 Possibility Theory

Possibility theory is explained in detail in [3], [6]. Possibility theory relies on the concept of possibility distributions.

Definition 1 *A possibility distribution is a function π that maps elements from a set of states U to a totally ordered scale $S = [0, 1]$, see [2]:*

$$\pi : U \to S.$$

The value $\pi(u)$ indicates the degree of possibility assigned to the state u. This function represents the agent's knowledge and helps differentiate between states that are more or less plausible. A state u is impossible if $\pi(u) = 0$, whereas it is considered fully possible if $\pi(u) = 1$.

Example 1 *Consider a weather forecasting scenario where an agent assesses the possibility of different weather conditions for tomorrow. Let $U = \{sunny, cloudy, rainy, stormy\}$ be the set of possible states. The agent assigns a possibility distribution π as follows:*
$\pi(sunny) = 1, \quad \pi(cloudy) = 0.7, \quad \pi(rainy) = 0.3, \quad \pi(stormy) = 0.$
Here, the agent is completely certain that a sunny day is possible ($\pi(sunny) = 1$), considers cloudy weather somewhat possible ($\pi(cloudy) = 0.7$), finds rain unlikely ($\pi(rainy) = 0.3$), and deems a storm impossible ($\pi(stormy) = 0$).

The basic measures of possibility theory model the possibility and necessity of events, see [2]. If the occurence of an event A is of interest possibility theory responds with both two measures calculated from the possibility distribution of A.

Definition 2 *The possibility of an event A is $\Pi(A) = \sup_{u \in A} \pi(u)$.*

The possibility is the maximum possibility of all the states of A. The possibility evaluates the extent to which A is consistent with π. Possibility measures satisfy the characteristic maxitivity property $\Pi(A \cup B) = \max(\Pi(A), \Pi(B))$.

Definition 3 *The necessity of an event A is $N(A) = \inf_{u \notin A} (1 - \pi(u))$.*

The necessity is the minimum of the inversed possibility value of all states which are not in A. The necessity evaluates the extent to which A is certainly implied by π. Necessity measures satisfy the characteristic minimization property $N(A \cap B) = \min(N(A), N(B))$. The minimization property expresses that to be certain about A and B, one has to be certain about A and to be certain about B.
The duality of the two measures is expressed as $N(A) = 1 - \Pi(A^c)$, where A^c is the complement of A.

Example 2 *Consider a possibility distribution π over a set of possible weather conditions: $U = \{sunny, cloudy, rainy\}$ with the values $\pi(sunny) = 1$, $\pi(cloudy) = 0.7$, and $\pi(rainy) = 0.3$.*

The possibility of the event "not rainy" $A = \{sunny, cloudy\}$ is given by:

$$\Pi(A) = \sup_{u \in A} \pi(u) = \max(\pi(sunny), \pi(cloudy)) = \max(1, 0.7) = 1.$$

The necessity of "not rainy" is computed as:

$$N(A) = \inf_{u \notin A} (1 - \pi(u)) = 1 - \pi(rainy) = 1 - 0.3 = 0.7.$$

This means that the event "not rainy" is fully possible but not completely necessary.

2.2 Formal foundations of Basic Possibilistic Logic

In the following the foundations of Basic Possibilistic Logic are layed out. First the syntax is introduced. Second the axioms of this logic are derived and third the inference rules are explained. The last part covers the semantic aspects of the logic.

2.2.1 Syntax

A basic possibilistic logic formula is represented as:

$$(a, \alpha)$$

where a is a classical logic formula (e.g., $p \to q$) and α is the certainty level (minimum necessity degree of a). It can be viewed as a lower bound of a necessity measure. The pair (a, α) is understood as $N(a) > \alpha$. We can interpret that the formula a is at least certain to degree α. The higher α, the stronger the belief in a. For example $(b \to f, 0.9)$ means that birds fly with certainty 0.9.

2.2.2 Axioms

The axioms of possibilistic logic are those of propositional logic. In the following the inference rules of basic possibilistic logic will be explained. The certainty weakening rule states that if a formula is valid with a high certainty level, it is also valid with any lower certainty level. If a formula holds with a certainty level α, it automatically implies that it holds with any lower level of certainty β because we are being less demanding, see equation 1.

$$\text{If } \beta \leq \alpha, \quad (a, \alpha) \vdash (a, \beta) \tag{1}$$

The modus ponens rules states that if a to b is true with certainty level α, and a is also true with certainty level α then b must be true with certainty level α, see equation 2.

$$(\neg a \vee b, \alpha), (a, \alpha) \vdash (b, \alpha), \quad \forall \alpha \in (0, 1] \tag{2}$$

In classical logic, the resolution rule allows us to combine two clauses that share a literal to eliminate that literal and infer a new clause. In possibilistic logic, the resolution rule preserves the certainty level α, because the inferred formula $b \vee c$ cannot be more certain than the least certain premise, see equation 3.

$$(\neg a \vee b, \alpha), \quad (a \vee c, \alpha) \vdash (b \vee c, \alpha) \tag{3}$$

The weakest link resolution rule extends the resolution rule by the concept of different certainty levels α and β. The weakest link resolution reflects the idea that the certainty of the conclusion cannot exceed the weakest certainty of the premises. If one of the premises is less certain, it becomes the bottleneck for the certainty of the conclusion, see equation 4.

$$(\neg a \vee b, \alpha), \quad (a \vee c, \beta) \vdash (b \vee c, \min(\alpha, \beta)) \tag{4}$$

2.2.3 Inference Rules

We can define a possibilistic knowledge base as a collection of possibilistic formulas, see equation 5.

$$\Gamma = \{(a_i, \alpha_i), i = 1, \ldots, m\} \tag{5}$$

Further we can define the α-cut of knowledge base, see equation 6. It only contains those formulas from the origin knowledge base with a certainty level of at least α. It has a nested structure where the α knwoledge base is a part of the β knowledge base if α is higher than β.

$$\Gamma_\alpha = \{(a_i, \alpha_i) \in \Gamma \mid \alpha_i \geq \alpha\} \tag{6}$$

A possibilistic inference (a, α) depends only on formulas with certainty levels $\geq \alpha$, see equation 7.

$$\Gamma \vdash (a, \alpha) \iff \Gamma_\alpha \vdash (a, \alpha) \tag{7}$$

If we want to infer from the knowledge base, we can only look on the formulas with a certainty of at least α. This means that a inference is only valid if it is valid with the α-cut of a knowledge base.
An important attribute of knowledge bases is the inconsistency level. It is the highest certainty level at which the knowledge base becomes inconsistent. The properties of the inconsistency level show that all formulas which have a higher certainty level than the inconsistency level are safe form inconsistency, see equation 8.

$$\text{inc-l}(\Gamma) = \max\{\alpha \mid \Gamma \vdash (\bot, \alpha)\} \tag{8}$$

2.2.4 Semantic Aspects

3 Differences from classical logic

TODO:
Handling of Inconsistencies

4 Relation to Artificial Intelligence

TODO:
Areas where this logic is beneficial
Examples of usage

5 Analysis and Development of Examples

In the following, examples showing the application of basic possibilistic logic are explained. First, the penguin problem is presented to provide a better understanding of the logic. Second, a new problem is introduced.

5.1 The Penguin Problem

One of the classical examples in non-monotonic reasoning is the penguin problem. Basic possibilistic logic provides a structured way to handle exceptions of general rules like "birds fly" by associating rules with certainty levels and defining a preference ordering over possible worlds.

5.1.1 Default Rules

The knowledge base of this example consists of three default rules that represent generic knowledge about birds and penguins:

$$\begin{aligned}
&d_1 : b \to f \quad (\text{"Birds fly"}),\\
&d_2 : p \to \neg f \quad (\text{"Penguins do not fly"}),\\
&d_3 : p \to b \quad (\text{"Penguins are birds"}).
\end{aligned}$$

In the following the goal is to assign certainty levels to these rules, to get a possibilistic knowledge base. In order to find this knowledge base, the following steps are necessary. First the constraints from the rules are constructed. Second, the possible worlds are ranked by the constraints. With the ranked worlds the certainty of the rules is calculated.

5.1.2 Constraints

From the default rules, a set of comparative possibility constraints can be constructed:

$$C_1 : b \wedge f >_\Pi b \wedge \neg f \quad \text{(birds fly)},$$
$$C_2 : p \wedge \neg f >_\Pi p \wedge f \quad \text{(penguins do not fly)},$$
$$C_3 : p \wedge b >_\Pi p \wedge \neg b \quad \text{(penguins are birds)}.$$

For example, C_1 states that the worlds where birds fly are more possible than worlds where birds do not fly. These constraints establish an ordering over possible worlds, which will be illuminated next.

5.1.3 Possible Worlds

The propositional language for this example involves three predicates: b (bird), p (penguin), and f (flies). The set of possible worlds, Ω, consists of all possible combinations of (b, f, p) represented by ω_i:

$$\Omega = \{\omega_0, \omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6, \omega_7\}.$$

Based on the constraints, these worlds are ranked to reflect which interpretations are more plausible.

World	b (Bird)	f (Flies)	p (Penguin)	d_1: $b \to f$	d_2: $p \to \neg f$	d_3: $p \to b$
ω_0	0	0	0			
ω_1	0	0	1		satisfied	violated
ω_2	0	1	0			
ω_3	0	1	1		violated	violated
ω_4	1	0	0	violated		
ω_5	1	0	1	violated	satisfied	satisfied
ω_6	1	1	0	satisfied		
ω_7	1	1	1	satisfied	violated	satisfied

Table 1: Truth values of bird (b), flies (f), and penguin (p) for each world ω_i, and whether each world satisfies or violates the default rules.

In table 1 each line represents a world with the combination of the three predicates and the satisfaction or violation of a default rule. In this table it can be observed that no world satisfies all rules, which highlight the core problem. The worlds ω_0 and ω_2 do not satisfy or violate any rule.

The first default rule, $d_1 : b \to f$ ("Birds fly"), increases the plausibility of worlds where birds fly ($b \wedge f$) compared to those where birds do not fly ($b \wedge \neg f$). This rule affects the ranking between worlds ω_6, ω_7 (where the bird flies) and ω_4, ω_5 (where the bird does not fly).

The second default rule, $d_2 : p \to \neg f$ ("Penguins do not fly"), prioritizes worlds where penguins do not fly ($p \wedge \neg f$) over those where penguins fly ($p \wedge f$). This affects the relative plausibility of ω_5, ω_1 (where the penguin does not fly) compared to ω_7 and ω_3 (where the penguin flies).
The third default rule, $d_3 : p \to b$ ("Penguins are birds"), ranks the worlds ω_5, ω_7 higher than ω_1, ω_3.
We can insert the worlds in which the rule holds into the constraints, as these worlds represent the rule:

$$\begin{aligned}
C_1 &: \max(\pi(\omega_6), \pi(\omega_7)) > \max(\pi(\omega_4), \pi(\omega_5)),\\
C_2 &: \max(\pi(\omega_5), \pi(\omega_1)) > \max(\pi(\omega_3), \pi(\omega_7)),\\
C_3 &: \max(\pi(\omega_5), \pi(\omega_7)) > \max(\pi(\omega_1), \pi(\omega_3)).
\end{aligned}$$

5.1.4 Ranking of the possible Worlds

To find out the correct order of the worlds, figure 1 visualizes the relationships. Only with the combination of the constraints it is possible to find out the correct ordering. The graph gives a hint about the possibility relations between the worlds.
For example the constraint C_1 says that either ω_6 (blue arrows) or ω_7 (red arrows) must be more possible than ω_4 and ω_5. The correct ordering is found if the worlds can be arranged in levels with the relations (arrows) while the arrows are only going down from a higher possibility level to a lower one. In this example an ordering can be found with color groups blue (C_1), orange (C_2) and yellow (C_3).

The ordering of the worlds threw the constraints results in:

$$\begin{aligned}
\pi(\omega_0) &= \pi(\omega_2) = \pi(\omega_6) = 1,\\
\pi(\omega_4) &= \pi(\omega_5) = \frac{2}{3},\\
\pi(\omega_1) &= \pi(\omega_3) = \pi(\omega_7) = \frac{1}{3}.
\end{aligned}$$

In this example the numerical representation is used for the different possibility levels of the worlds. The worlds ω_0 and ω_1 are in the highest possibility level, as no constraints grades the worlds down.

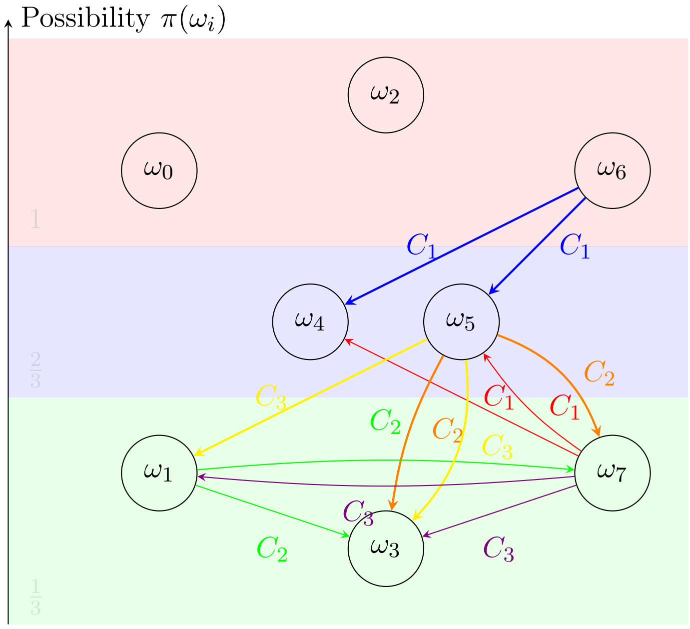

Figure 1: Possibility of the worlds ω_i and possibility relation between the worlds represented by the colored arrows

5.1.5 Creating Knowledge Base

We calculate the necessity degrees from the possibility of the worlds as follows with the duality relationship of possibility and necessity:

$$\begin{aligned}
N(\neg p \vee \neg f) &= \min\{1 - \pi(\omega) \mid \omega \models p \wedge f\} \\
&= \min(1 - \pi(\omega_3), 1 - \pi(\omega_7)) = \frac{2}{3} \\
N(\neg b \vee f) &= \min\{1 - \pi(\omega) \mid \omega \models b \wedge \neg f\} \\
&= \min(1 - \pi(\omega_4), 1 - \pi(\omega_5)) = \frac{1}{3} \\
N(\neg p \vee b) &= \min\{1 - \pi(\omega) \mid \omega \models p \wedge \neg b\} \\
&= \min(1 - \pi(\omega_1), 1 - \pi(\omega_3)) = \frac{2}{3}
\end{aligned}$$

Thus, we obtain the possibilistic knowledge base:

$$K = \{(\neg p \vee \neg f, \frac{2}{3}), (\neg p \vee b, \frac{2}{3}), (\neg b \vee f, \frac{1}{3})\}.$$

The possibilitic knowledge base is visualized in figure 2.

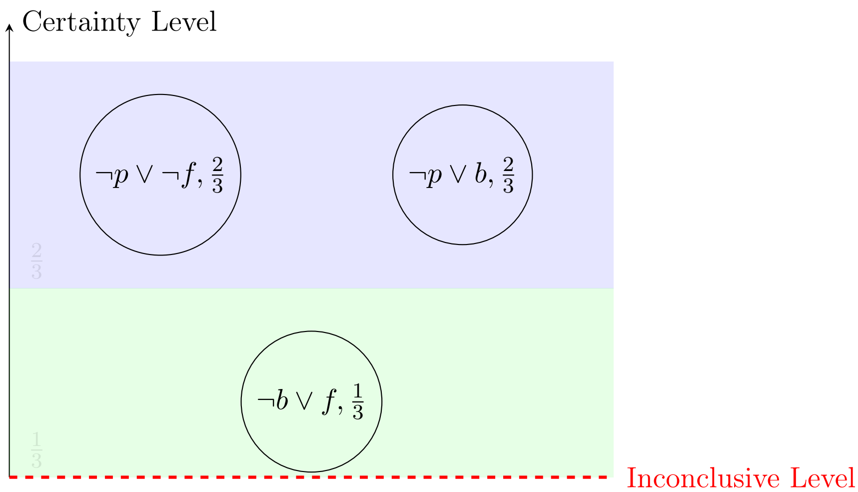

Figure 2: Possibilistic Knowledge Base of the Penguin Problem with the elements arranged over the certainty

5.1.6 Reasoning with Uncertain Information

In this part we can find out if Tweety flies by introducing information about Tweety. First we introduce that Tweety is a bird:

$$F = \{(b, 1)\} \quad \text{(Tweety is a bird with certainty 1).}$$

From the rule $(\neg b \vee f, 1/3)$ in the knowledge base K, we can infer:

$$(\neg b \vee f, 1/3) \quad \text{and} \quad (b, 1) \vdash (f, \min(1, 1/3)) = (f, 1/3).$$

Thus, we conclude that Tweety flies with a certainty level of $\frac{1}{3}$. We can see the result graphically in figure 3. The violet arrow symbolizes the inference with the weakest link resolution rule.

If new information states that Tweety is also a penguin:

$$F = \{(b, 1), (p, 1)\} \quad \text{(Tweety is a penguin with certainty 1),}$$

then, using $(\neg p \vee \neg f, 2/3)$, we derive:

$$(\neg p \vee \neg f, 2/3) \quad \text{and} \quad (p, 1) \vdash (\neg f, \min(1, 2/3)) = (\neg f, 2/3).$$

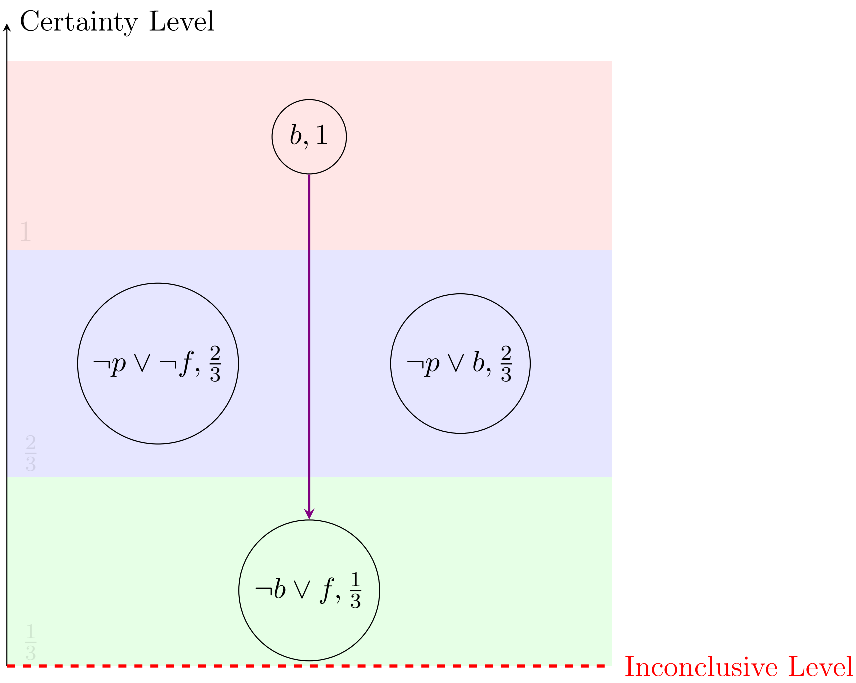

Figure 3: Possibilistic Knowledge Base of the Penguin Problem with first information about Tweety

Since we now have conflicting conclusions:

$$(\neg f, 2/3) \quad \text{and} \quad (f, 1/3) \vdash (\perp, \min(2/3, 1/3)) = (\perp, 1/3),$$

the system detects an inconsistency of level $\frac{1}{3}$. Given the principle that higher certainty information takes advantage, the inference that Tweety flies is drowned, leading to the final conclusion:

$$K \cup F \vdash (\neg f, 2/3),$$

which means *"Tweety does not fly"* with a certainty of $\frac{2}{3}$.
The results are visualized in the figure 4. It can be seen that the inconclusive level is raised to a level where above no conflict is between the information.

The process is visualized in the flowchart 5.

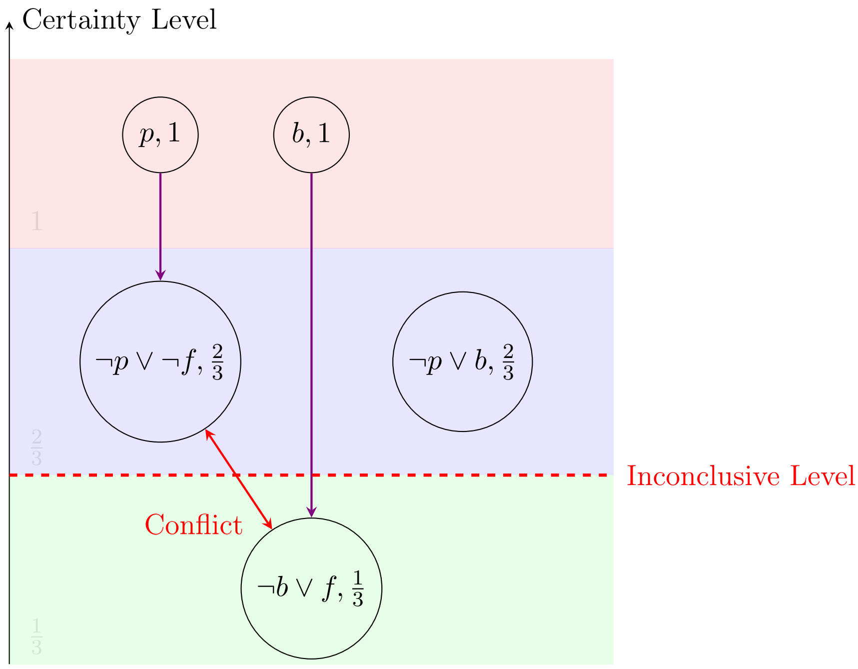

Figure 4: Possibilistic Knowledge Base of the Penguin Problem with new information about Tweety

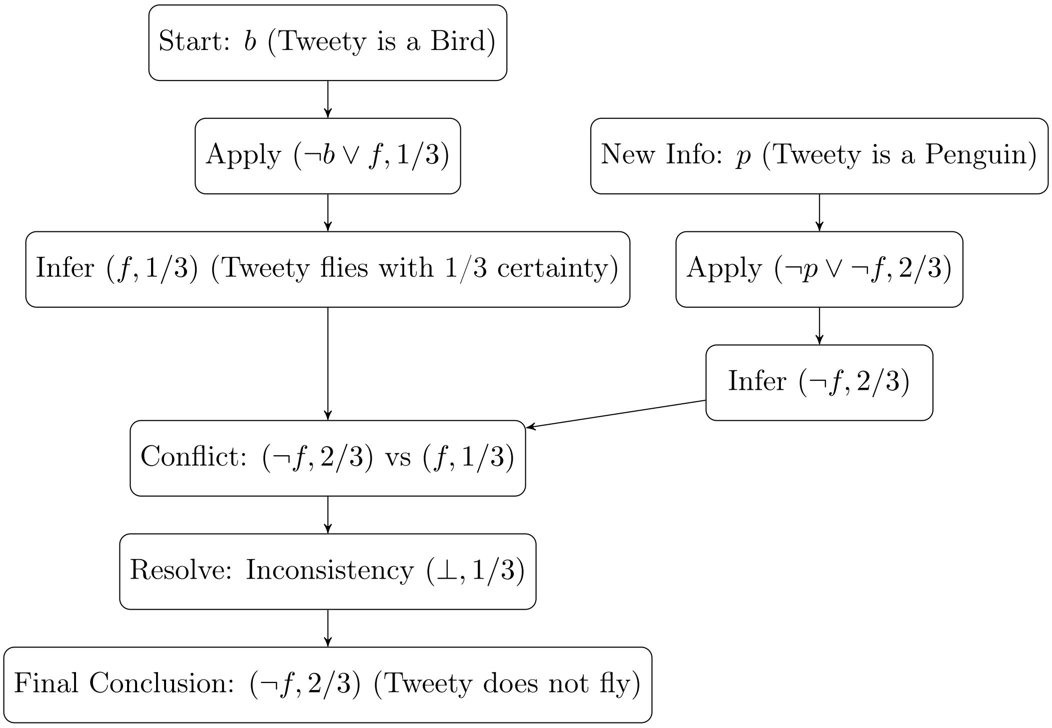

Figure 5: Ranking of the worlds

5.2 New Problem

TODO:
More complex Problem
Medical Problem

6 Discussion and Conclusion

References

[1] Didier Dubois, Jérôme Lang, and Henri Prade. A brief overview of possibilistic logic. In Rudolf Kruse and Pierre Siegel, editors, *Symbolic and Quantitative Approaches to Uncertainty: European Conference ECSQAU, Marseille, France, October 15-17, 1991. Proceedings*, volume 548 of *Lecture Notes in Computer Science book series (LNCS)*, pages 53–57, Marseilles, France, October 1991. Springer Berlin Heidelberg.

[2] Didier Dubois and Henri Prade. Possibilistic logic — an overview. In Jörg H. Siekmann, editor, *Computational Logic*, volume 9 of *Handbook of the History of Logic*, pages 283–342. North-Holland, 2014.

[3] Didier Dubois and Henri Prade. Possibility theory. In *Granular, Fuzzy, and Soft Computing*, pages 859–876. Springer, 2023.

[4] Joseph Y. Halpern. An analysis of first-order logics of probability. *Artificial Intelligence*, 46(3):311–350, 1990.

[5] L.A. Zadeh. Fuzzy sets. *Information and Control*, 8(3):338–353, 1965.

[6] L.A Zadeh. Fuzzy sets as a basis for a theory of possibility. *Fuzzy Sets and Systems*, 1(1):3–28, 1978.

Statement

Ich erkläre, dass ich die schriftliche Ausarbeitung zum Seminar selbstständig und ohne unzulässige Inanspruchnahme Dritter verfasst habe. Ich habe dabei nur die angegebenen Quellen und Hilfsmittel verwendet und die aus diesen wörtlich oder sinngemäß entnommenen Stellen als solche kenntlich gemacht. Die Versicherung selbstständiger Arbeit gilt auch für enthaltene Zeichnungen, Skizzen oder graphische Darstellungen. Die Ausarbeitung wurde bisher in gleicher oder ähnlicher Form weder derselben noch einer anderen Prüfungsbehörde vorgelegt und auch nicht veröffentlicht. Mit der Abgabe der elektronischen Fassung der endgültigen Version der Ausarbeitung nehme ich zur Kenntnis, dass diese mit Hilfe eines Plagiatserkennungsdienstes auf enthaltene Plagiate geprüft werden kann und ausschließlich für Prüfungszwecke gespeichert wird.

....

(Place, Date) (Signature)